

Changes in Memory, Thinking, and Focus (Chemo Brain)

Some people with cancer notice that they can't think as clearly as they used to. They may have trouble remembering things, focusing, finishing tasks, or learning something new. These are symptoms of **cognitive impairment**, also known as “**chemo brain**” or “**brain fog**.”

Memory, thinking, and focus changes typically go away over time. For most people, these thinking changes are not signs of a more serious condition like Alzheimer's disease.

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What causes changes in memory, thinking, and focus?

It is not clear what causes changes in thinking. But studies show that some cancer treatments can cause changes in how your brain works.

While this side effect is often called “chemo brain,” people with cancer can have memory, thinking, and focus changes for other reasons. Changes can happen in people with cancer who have not been treated, gotten other treatments, or have been off treatment for a while.

Cancer treatments that may increase the risk of changes in thinking include:

- Chemotherapy
- Surgery
- Radiation
- Immunotherapy
- Hormone therapy
- Targeted drug therapies
- Other medicines used as part of treatment, such as steroids, anti-nausea, or pain medicines

Other things can increase the risk of changes in thinking or make these changes worse, such as:

- The cancer itself, such as brain tumors
- Other conditions or illnesses, such as diabetes or high blood pressure
- Symptoms of cancer or cancer treatment, such as tiredness, pain, or trouble sleeping
- Infection
- Stress, anxiety, or depression
- Low red blood cell counts (anemia)
- Hormone changes, such as being post-menopausal
- Not eating enough or getting the right nutrients in your diet
- Older age
- Being weak or frail
- Using alcohol or other substances that can affect your brain

Most of these cause short-term problems that get better as the cause is treated or goes away on its own. Others can lead to longer-lasting changes in thinking unless the cause is treated.

Symptoms of changes in memory, thinking, and focus

Changes in thinking may not be easy for other people to see. People in your life might not even notice anything. Still, if you have these changes, you will most likely know it.

If you have changes in thinking you may have trouble:

- Remembering things that you can usually recall (memory lapses)
- Concentrating (not being able to focus on what you're doing, having a short

- attention span, or losing track of what's going on around you)
- Remembering details like names, dates, and events
- Coming up with ideas, planning activities, and making decisions (executive dysfunction)
- Losing track of one task when you're trying to do more than one thing at a time, like answering the phone while cooking (trouble multi-tasking)
- Learning new things
- Taking longer to finish tasks (feeling disorganized and thinking more slowly)
- Finding words (unable to find the right words to finish a sentence)

Changes in thinking might interfere with your usual activities like school, work, hobbies, or social get-togethers. These activities might take more mental effort than usual and be tiring.

Changes in thinking can happen at any time when you have cancer. It can happen before, during, and after cancer treatment.

For most people, these changes only last a short time. Other people can have long-term or delayed symptoms. When changes in thinking start, how long they last and how much trouble they cause may be different for each person.

Treatment for thinking changes

There are some things that can be done to help with changes in thinking.

- **Exercise:** Be as active as you can. Regular physical activity is good for your brain. It can improve your thinking and focus. It can also help you deal with stress and make you feel more alert and less tired.
- **Meditation:** Meditation can help you relax, clear your mind, and focus on the present time.
- **Movement therapy:** Practicing yoga, tai chi, or qi gong might improve your focus and thinking.
- **Focused rehabilitation:** Some types of therapy can help improve focus, thinking, and memory. These types of treatment might also help you organize your usual activities or improve your job skills.

Talk to your cancer care team about these and other options that may help you cope with changes in thinking.

Tips to manage changes in memory, thinking, and focus

As you cope with these changes, you might want to:

- **Use a daily planner, notebook, notes, or your smartphone for reminders.** Keeping everything in one place makes it easier to find the reminders you may need. Keep track of appointments, to-do lists, important dates, phone numbers and addresses, and meeting notes. You might even want to write down movies you'd like to see or books you'd like to read.
- **Set up routines and follow them.** Try to stick to the same daily schedule.
- **Pick a certain place to keep items that often get lost,** like your keys or the remote. Put each item in its place after you use it.
- **Try to focus on one thing at a time.**
- **Do your hardest tasks when you have the most energy.**
- **Exercise your brain.** Do puzzles, solve word games, paint pictures, play an instrument, or learn a new hobby.
- **Try to get enough rest and sleep.**
- **Ask for help when you need it.** Having friends and loved ones help with daily tasks can cut down on distractions and help you save mental energy.
- **Track your memory problems.** Keep a diary or log of when you notice problems and what's going on at the time. Keeping track of medicines you take, the time of day, and where you are might help you figure out what affects your memory. This record can also help you determine the best times to plan important tasks or appointments.

Tell others what is happening

Remember, you probably notice your problems much more than others do. It might help to talk to your family, friends, and cancer care team about your memory, thinking, and focus changes. Let them know what you're going through. You may feel relieved to tell people about the problems you are having.

A change in thinking, memory, or focus is a side effect you can learn to manage. Even though this side effect may not be easy to see, your family and friends might be able to help you. For instance, they might notice that when you're rushed, you have more trouble finding things.

Tell your friends and family members what they can do to help. Their support and

understanding can help you relax and make it easier for you to focus and process information.

When to talk with your doctor or cancer care team

Let your cancer care team know if you have any changes in your memory, thinking, or focus. You might notice these changes soon after diagnosis, during treatment, or after treatment ends. Try to explain:

- What you are feeling
- If there are certain times of day you notice these changes the most
- What seems to make the changes better or worse
- How these changes are affecting your life
- Other symptoms that seem to be related, such as fatigue, trouble sleeping, pain, or stress

Keeping a diary or log can help you keep track of when you have changes in thinking and what seems to affect them. Take your diary or log with you when you see your cancer care team.

Other ways to make the most of your visit:

- **Write down your questions and take them with you.** This will help you remember what you wanted to ask.
- **Bring a list of all the medicines you take**, including any prescriptions, over-the-counter medicines, herbs, vitamins, and supplements.
- **Take a friend or family member with you.** They can help you remember what you want to ask and keep track of what's said during the visit. Your friend or relative can also describe the changes they've seen.

Tell your cancer care team how much of a problem the changes in thinking are for you. Are they keeping you from what you want to do? From going to school or work? Are you having trouble taking care of your family? Are you able to take care of yourself?

Also let your cancer care team know if you are feeling anxious or depressed due to the changes in thinking. Talking to a mental health professional might help you deal with these feelings.

You might also want to ask about focused rehabilitation to help you manage specific

changes in focus, thinking, and memory.

Questions to ask about changes in memory, thinking, and focus

Here are some questions you may want to ask your doctor:

- Am I at risk for changes in memory, thinking, or focus?
- When might these changes happen? How long are they likely to last?
- Are there other medical problems or medicines that could be causing my symptoms?
- Is there treatment for my symptoms?
- What can I do for my changes in memory, thinking, and focus?
- Is there anything I can do to help prevent or decrease these changes?
- Should I see a rehabilitation specialist or mental health professional? Can you recommend one?

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